A counterexample to the Noferini–Poloni conjecture for Hurwitz stable matrices

1 Statement

For $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$ let

$$\mathcal{H}_n(\mathbb{F}) = \{B \in \mathbb{F}^{n\times n} : \ \operatorname{Re}\lambda \le 0 \text{ for every } \lambda \in \sigma(B)\},$$

and for $A \in \mathbb{R}^{n\times n}$ put

$$d_{\mathbb{F}}(A) = \min_{B\in\mathcal{H}_n(\mathbb{F})} \|A - B\|_F, \qquad \|M\|_F^2 = \sum_{i,j} |M_{ij}|^2.$$

No semisimplicity condition is imposed on eigenvalues on the imaginary axis. Since $\mathcal{H}_n(\mathbb{R}) \subset \mathcal{H}_n(\mathbb{C})$, we always have $d_{\mathbb{C}}(A) \le d_{\mathbb{R}}(A)$; the Noferini–Poloni conjecture [1] asserts equality for every real A. Equivalently, the complex problem always has a real global minimizer.

Theorem 1. *Let*

$$A = \begin{pmatrix} -16 & 3 & 0 \\ 0 & 0 & 4 \\ 48 & 1 & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3}.$$

Then $d_{\mathbb{R}}(A) = \sqrt{10} \approx 3.1623$, *whereas* $d_{\mathbb{C}}(A) \le \frac{\sqrt{93781}}{100} \approx 3.0624 < \sqrt{10}$. *In particular* $d_{\mathbb{R}}(A) \neq d_{\mathbb{C}}(A)$, *and the conjecture is false for* $n = 3$.

2 The real problem: $d_{\mathbb{R}}(A) = \sqrt{10}$

Lemma 1. *If* $B \in \mathcal{H}_3(\mathbb{R})$, *then* $\det B \le 0$.

Proof. The characteristic polynomial of B is real, so non-real eigenvalues occur in conjugate pairs. As $n = 3$ is odd, B has at least one real eigenvalue $\lambda \le 0$. If all eigenvalues are real, they are all ≤ 0 and $\det B$ is a product of three nonpositive numbers, hence ≤ 0. Otherwise the other two eigenvalues are $\mu, \bar{\mu}$ and $\det B = \lambda|\mu|^2 \le 0$. □

Lemma 2 (Eckart–Young)**.** *For* $A \in \mathbb{C}^{n\times n}$, $\min\{\|A - S\|_F : \det S = 0\} = \sigma_{\min}(A)$.

Proposition 1. *For the matrix* A *of Theorem 1,* $d_{\mathbb{R}}(A) = \sqrt{10}$.

Proof. Lower bound. Expanding along the second row, $\det A = -4 \det \begin{pmatrix} -16 & 3 \\ 48 & 1 \end{pmatrix} = -4(-160) = 640 > 0$. The columns of A are $(-16, 0, 48)^T$, $(3, 0, 1)^T$, $(0, 4, 0)^T$; they are pairwise orthogonal with squared norms $2560, 10, 16$. Hence $A^T A = \operatorname{diag}(2560, 10, 16)$ and $\sigma_{\min}(A) = \sqrt{10}$.

Let $B \in \mathcal{H}_3(\mathbb{R})$. The function $t \mapsto \det(A + t(B - A))$ is continuous on $[0, 1]$, equals $640 > 0$ at $t = 0$, and is ≤ 0 at $t = 1$ by Lemma 1. So there is $t_* \in (0, 1]$ with $S := A + t_*(B - A)$ singular, and by Lemma 2

$$\|A - B\|_F \;\geq\; t_*\|B - A\|_F = \|A - S\|_F \;\geq\; \sigma_{\min}(A) = \sqrt{10}.$$

Thus $d_{\mathbb{R}}(A) \geq \sqrt{10}$.

Attainment. Let $B_{\mathbb{R}} = A - \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} e_2^T = \begin{pmatrix} -16 & 0 & 0 \\ 0 & 0 & 4 \\ 48 & 0 & 0 \end{pmatrix}$. Then $\det(zI - B_{\mathbb{R}}) = z^2(z + 16)$, so $B_{\mathbb{R}} \in \mathcal{H}_3(\mathbb{R})$, and $\|A - B_{\mathbb{R}}\|_F^2 = 3^2 + 1^2 = 10$. Hence $d_{\mathbb{R}}(A) = \sqrt{10}$. □

3 A closer complex stable matrix

Define $B \in \mathbb{C}^{3\times 3}$ by

$$B = \begin{pmatrix} -16.17 - 0.03\,\mathrm{i} & 1.09 - 0.09\,\mathrm{i} & -0.38 - 1.84\,\mathrm{i} \\ -0.03\,\mathrm{i} & -0.52 + 0.68\,\mathrm{i} & 3.30 - 0.10\,\mathrm{i} \\ 48.03 - 0.01\,\mathrm{i} & 0.36 - 0.13\,\mathrm{i} & -0.28 - 0.65\,\mathrm{i} \end{pmatrix}.$$

All entries are exact decimals.

Proposition 2. $\|A - B\|_F^2 = \dfrac{93781}{10000} = 9.3781 < 10.$

Proof. $E = B - A$ has entries

$$E = \begin{pmatrix} -0.17 - 0.03\mathrm{i} & -1.91 - 0.09\mathrm{i} & -0.38 - 1.84\mathrm{i} \\ -0.03\mathrm{i} & -0.52 + 0.68\mathrm{i} & -0.70 - 0.10\mathrm{i} \\ 0.03 - 0.01\mathrm{i} & -0.64 - 0.13\mathrm{i} & -0.28 - 0.65\mathrm{i} \end{pmatrix},$$

and the squared moduli of its entries, row by row, sum to $(0.0298 + 3.6562 + 3.5300) + (0.0009 + 0.7328 + 0.5000) + (0.0010 + 0.4265 + 0.5009) = 9.3781$. □

Proposition 3. *B is Hurwitz stable; in fact every eigenvalue has negative real part.*

Proof. We use the Lyapunov criterion: if $P = P^* \succ 0$ and $Q := -(B^*P + PB) \succ 0$, then for any eigenpair $Bv = \lambda v$, $v \neq 0$, $0 < v^*Qv = -2\,\mathrm{Re}(\lambda)\, v^*Pv$, so $\mathrm{Re}\,\lambda < 0$. Take

$$P = \begin{pmatrix} 1161764 & 107585 + 380958\,\mathrm{i} & 390891 - 68804\,\mathrm{i} \\ 107585 - 380958\,\mathrm{i} & 136577 & 13936 - 134452\,\mathrm{i} \\ 390891 + 68804\,\mathrm{i} & 13936 + 134452\,\mathrm{i} & 135688 \end{pmatrix}.$$

Its leading principal minors are 1161764, 1966711839, 67011874751, all positive, so $P \succ 0$. A direct computation in exact rational arithmetic gives

$$Q = \begin{pmatrix} \frac{9769}{10} & -\frac{2673}{100} - \frac{1853}{100}\mathrm{i} & \frac{125}{4} - \frac{137}{25}\mathrm{i} \\ -\frac{2673}{100} + \frac{1853}{100}\mathrm{i} & \frac{50041}{50} & -\frac{7}{20} + \frac{7}{100}\mathrm{i} \\ \frac{125}{4} + \frac{137}{25}\mathrm{i} & -\frac{7}{20} - \frac{7}{100}\mathrm{i} & \frac{24893}{25} \end{pmatrix},$$

whose leading principal minors are (exact rationals, approximately) 976.9, 9.766×10^5, 9.715×10^8, all positive; so $Q \succ 0$. □

Remark 1 (Independent check). The characteristic polynomial of B is $z^3 + \frac{1697}{100}z^2 + \big(\frac{15311}{500} + \frac{221389}{2500}\mathrm{i}\big)z - \frac{112253477}{1000000} + \frac{2518139}{40000}\mathrm{i}$, with roots $\approx -16.4746 + 5.4536\,\mathrm{i}$, $-0.3420 - 2.4072\,\mathrm{i}$, $-0.1534 - 3.0464\,\mathrm{i}$.

4 Proof of Theorem 1

By Propositions 3 and 2, $B \in \mathcal{H}_3(\mathbb{C})$ and $\|A-B\|_F < \sqrt{10}$, so $d_{\mathbb{C}}(A) \le \|A-B\|_F = \frac{\sqrt{93781}}{100} < \sqrt{10}$. By Proposition 1, $d_{\mathbb{R}}(A) = \sqrt{10}$. Hence $d_{\mathbb{C}}(A) < d_{\mathbb{R}}(A)$, and no real matrix in $\mathcal{H}_3(\mathbb{C})$ is a global minimizer of $\|A - \cdot\|_F$ over $\mathcal{H}_3(\mathbb{C})$. □

5 Remarks

Remark 2 (Mechanism). A real stable 3×3 matrix has $\det \le 0$ (Lemma 1), while $\det A > 0$, so every real competitor must cross the singular matrices, which costs at least $\sigma_{\min}(A)$. Over $\mathbb{C}$ there is no sign constraint on det for stable matrices (e.g. $\mathrm{diag}(\mathrm{i}, \mathrm{i}, -1)$ has $\det = 1$), and the spectrum need not be closed under conjugation. Numerically the optimal B has spectrum $\{-16.2 + 5.4\mathrm{i},\ -2.7\mathrm{i},\ -2.7\mathrm{i}\}$ up to rounding; the minimum of the complex problem is about 3.0054 (a local optimization over the unitary group, not needed for the proof).

Remark 3 (Role of the closed half-plane). The real minimizer $B_{\mathbb{R}}$ has the double eigenvalue 0 with a single Jordan block, so it lies on the boundary of $\mathcal{H}_3(\mathbb{R})$ and is not semisimple, which is allowed by the definition of $\mathcal{H}_n$ above. The lower bound $d_{\mathbb{R}}(A) \ge \sqrt{10}$ does not use $B_{\mathbb{R}}$.

Remark 4 (Related work). A disproof of the conjecture for Hurwitz stable matrices has also been reported by Noferini and Zhukov [2], via a family of 3×3 matrices with orthogonal columns and the same determinant-sign obstruction. The example above was constructed and verified independently.

References

[1] V. Noferini and F. Poloni, *Nearest Ω-stable matrix via Riemannian optimization*, Numer. Math. 148(4):817–851, 2021.

[2] V. Noferini and M. Zhukov, *Matrix nearness problems: Do real inputs admit real solutions?*, arXiv:2609.14828, 2026.